Finetuning VoxCPM for Speech Control: Integrating Non-Verbal Vocalization

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1 Overview

This report concerns the control of delivery in synthetic speech: the ability to specify how an utterance is spoken rather than only what is said. It has three purposes. Section 1 describes VoxCPM2, the model this project has adopted for Khmer, and reports a measurement of the control it already provides. Section 2 defines the categories of speech control and distinguishes them from one another. Section 3 reviews the published work addressing each category and identifies the one this project will pursue. Section 4, the methodology, is reserved.

1.1 Model Description

VoxCPM2 is an open-weight text-to-speech model released by OpenBMB under the Apache 2.0 licence. It has 2.29 billion parameters and is tokenizer-free: rather than mapping speech onto a discrete codebook, it predicts continuous latent vectors, one for every four-frame patch of audio.

This property has a direct bearing on low-resource languages. Systems built on discrete audio codebooks depend on the codebook having been fitted to the target language during pre-training, and where it has not, synthesis degrades in a way that fine-tuning does not readily repair. In the four-model comparison conducted for this project, Fish Audio S2-Pro failed in exactly that manner, returning a median character error rate of 78.01 per cent on Khmer. Over the same fixed set of one hundred sentences VoxCPM2 returned 2.47 per cent, against 8.28 per cent for Higgs TTS 3 and 25.12 per cent for Meta MMS. VoxCPM2 was selected on that evidence.

A second property follows from the input side. The model reads raw UTF-8 bytes through a 73,448-entry tokenizer and is given no language identifier, inferring the language from the script. Khmer consequently requires no grapheme-to-phoneme conversion, no pronunciation lexicon and no word segmenter. None of these components exists for Khmer in a form suitable for production use, and assembling them ordinarily accounts for the largest share of effort in a low-resource text-to-speech project.

1.2 System Architecture

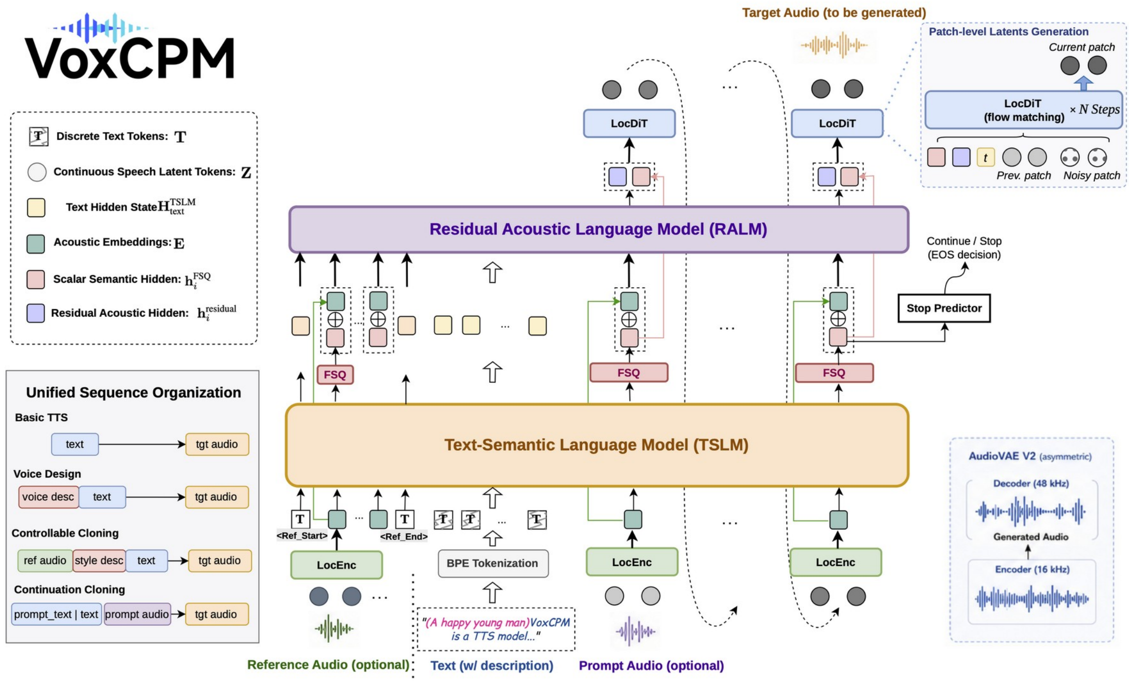

Figure 1: Overview of VoxCPM's Architecture

The generation path comprises the stages set out in Table 1. A backbone language model emits one latent vector per audio patch; a residual language model refines it; a local diffusion transformer converts the refined latent into acoustic features under a conditional flow-matching objective; and a variational autoencoder decodes those features to a waveform, accepting features derived at 16 kHz and emitting audio at 48 kHz. A separate local encoder, not shown, maps a reference recording into the same latent space and is the mechanism by which zero-shot voice cloning is performed.

Table 1. The VoxCPM2 generation path. Dimensions and layer counts are taken from `config.json` in the `openbmb/VoxCPM2` release.

Stage	Configuration	Output
Byte-level tokenizer	vocabulary 73,448; no grapheme-to-phoneme stage, no language identifier	token sequence
MiniCPM4 backbone LM	2048 dimensions, 28 layers; grouped-query attention, 16 query and 2 key-value heads; LongRoPE to 32k	one latent vector per audio patch
Residual LM	8 layers, no positional encoding	refined latent
Local DiT	1024 dimensions, 12 layers; conditional flow matching; Euler solver, guidance scale 2.0, 10 steps at inference	64-dimensional acoustic features, 4 frames per patch
AudioVAE V2	encoder at 16 kHz, decoder at 48 kHz	waveform

Table 2. Architecture parameters bearing on training and control.

Parameter	Value	Consequence
`patch_size`	4	One language-model step spans four autoencoder frames.
`feat_dim`	64	Width of the latent the diffusion transformer predicts.
Frame rate	25 fps	One second of audio occupies 6.25 language-model positions.
Encoder rate	16 kHz	Training audio must be supplied at 16 kHz; the validator rejects other rates.
Decoder rate	48 kHz	Bandwidth extension is internal; 48 kHz training data is not required.
`inference_cfg_rate`	2.0	Default classifier-free guidance scale, exposed as `--cfg-value`.

Two of these properties bear on the remainder of the report.

The first is the identity of the acoustic decoder. The local diffusion transformer is a conditional flow-matching model. The methods reviewed in Section 3.3 for controlling non-verbal vocalization were developed for, and evaluated on, models of this class. They are therefore applicable to VoxCPM2 without alteration of the underlying training objective, which is not true of methods developed for discrete-codec systems.

The second concerns the treatment of the text field during training. The loss mask constructed in the data packer is zero at every text position:

```
# voxcpm/training/packers.py, process_tts_data
loss_mask = cat([zeros(text_length), ones(audio_length), zeros(1)])
#                ^^^^^^^^^^^^^^^^^^ zero at every text position
```

No token in the text field is ever a prediction target. The field operates purely as a conditioning channel, and its contents may therefore be extended with tags, markers or descriptive text without perturbing the objective the model is trained under. Both control mechanisms described below exploit this, and any mechanism added later would do the same.

1.3 Built-in Control Mechanisms

VoxCPM2 provides two mechanisms for influencing delivery. They differ in the interval over which they apply, and that difference is developed in Section 2.

The first is a parenthetical description placed before the text, which characterises the utterance as a whole:

```
model.generate(
    text="(speaking quickly, a high-pitched voice)"
         "ថ្ងៃនេះអាកាសធាតុល្អណាស់។")
```

The second is a bracketed tag placed inline, which marks a single event at one position in the text:

```
model.generate(text="ខ្ញុំគិតថាមិនអីទេ [laughing] ប៉ុន្តែ...")
```

The documented tag inventory is:

```
[laughing]   [laughter]   [sigh]   [Uhm]   [Shh]
[Question-ah] [Question-ei] [Question-en] [Question-oh]
[Surprise-wa] [Surprise-yo] [Dissatisfaction-hnn]
```

The model documentation describes both mechanisms with reference to Chinese and English. It makes no statement about their behaviour in other languages, and the training corpus composition is not published in sufficient detail to infer one. Their efficacy on Khmer is therefore an empirical question, which the next section addresses for the first mechanism.

1.4 Measurement of Parenthetical Prosodic Control on Khmer

The parenthetical mechanism was evaluated on the unmodified model, using eight sentences drawn from the project's fixed Khmer evaluation set. For each of four prosodic axes, three prompts were written to span the axis (Table 3). Every sentence was synthesised under every prompt at three random seeds, giving 96 generations, and the median taken within each cell of the design. Acoustic measurement follows the definitions used elsewhere in the project: median fundamental frequency for pitch, its standard deviation in semitones for variation, root-mean-square level for energy, and Khmer characters per second for rate.

Table 3. Prompts used to span each prosodic axis.

Axis	Level 0	Level 1	Level 2
Pitch	*(a low-pitched voice)*	*(a normal-pitched voice)*	*(a high-pitched voice)*
Energy	*(speaking softly, quietly)*	*(speaking at a normal volume)*	*(speaking loudly)*
Rate	*(speaking slowly)*	*(speaking at a normal pace)*	*(speaking quickly)*
Variation	*(a flat, monotone delivery)*	*(a normal delivery)*	*(a lively, expressive delivery)*

Table 4. Response of the unmodified model to a parenthetical prompt on Khmer. The corpus bound is the separation between the low and high bands of this project's hand-labelled Khmer corpus, and represents an upper limit on what fine-tuning against that corpus could teach.

Axis	Unit	Level 0	Level 1	Level 2	Change	Seed s.d.	Corpus bound	ρ	p
Pitch	Hz	147.80	202.81	254.16	+106.37	28.78	28.27	+0.759	0.0002
Energy	dBFS	-20.38	-16.30	-15.89	+4.50	5.69	5.81	+0.605	0.0022
Rate	char/s	14.67	15.84	16.96	+2.29	1.58	6.57	+0.590	0.0035
Variation	st	4.55	4.06	3.92	-0.63	0.68	1.71	-0.273	0.2039

ρ is the Spearman rank correlation between prompt level and measured value; p is a permutation test over 10,000 relabellings. Seed s.d. is the mean run-to-run standard deviation within a cell.

Three findings follow.

Pitch is controlled reliably. The separation between the extreme prompts is 106.37 Hz with a rank correlation of +0.759 ($p = 0.0002$). This exceeds the corpus bound of 28.27 Hz by a factor of approximately four. The capability a fine-tune on the project's own labelled data could add to this axis is therefore negative.

Energy and **Speaking Rate** are controlled in aggregate but not per generation. Both show significant rank correlations (+0.605 and +0.590), but the energy effect of 4.50 dB is smaller than the 5.69 dB standard deviation observed across seeds within a single cell. The ordering of the levels is dependable; the value of any individual synthesis is not.

Pitch Variation is not controlled. The correlation is negative (-0.273) and not significant ($p = 0.20$). Prompts requesting a lively delivery produced marginally less pitch movement than prompts requesting a monotone one, which is consistent with the axis being unaddressed rather than inverted.

The behaviour of the inline tags on Khmer was not measured, and no claim is made about it here. Establishing whether they fire at all on Khmer text is a prerequisite for the work Section 3.5 selects.

2 Speech Control

2.1 Definition and Scope

The term expressive control is used loosely in the literature to denote any influence over delivery not exercised through the choice of words. It subsumes several capabilities that differ in what they describe, over what interval they apply, and in the effort each requires to implement. This section separates the three that concern this project, and names three further categories that are adjacent to them.

One distinction cuts across all of them and is used throughout what follows. A global attribute is a property of a whole utterance, or of a long span of one: it has no onset and no offset, and it is realised in every frame. A local event is bounded: it begins, occupies a short interval, and ends, and it is associated with a specific position in the text. The two are not merely different in duration. A global attribute must influence frames whose acoustic content is already largely determined by the surrounding speech, whereas a local event is the only thing that determines the frames it occupies. The consequences for training are taken up in Section 3.5.

2.2 Prosody

Prosody comprises the suprasegmental properties of speech, that is, those carried above the level of the individual sound. It is what remains when the identity of the words is set aside: the rate at which they are spoken, the pitch at which they are set, the loudness with which they are projected, the degree of pitch movement, and the placement of pauses.

Table 5. Prosodic dimensions and their acoustic correlates.

Dimension	Acoustic correlate	Measurement used here
Speaking rate	phones or syllables per unit time	Khmer characters per second
Pitch register	fundamental frequency	median F0, hertz
Pitch variation	F0 range and contour movement	F0 standard deviation, semitones
Loudness	signal energy	root-mean-square level, dBFS
Phrasing	pause placement and duration	inter-pausal unit statistics

The sentence *I never said she stole my money* spoken at three syllables per second and at six differs in rate alone. Spoken with a median fundamental frequency of 120 hertz and of 240 hertz, it differs in register. Spoken with an F0 standard deviation near zero it is heard as mechanical, and with a range of six semitones as engaged; that is variation. In each case the words, and the meaning they carry, are unchanged.

Prosody is a global attribute. It is the category most extensively treated in the literature, and, as Section 1.4 established, the one VoxCPM2 already addresses on Khmer.

2.3 Emotion

Emotion denotes the affective state conveyed by the delivery. The standard corpora encode it as a small closed set of categories, most commonly neutral, happy, angry, sad and surprised.

Emotion is realised through prosody together with voice quality, but it does not reduce to a prosodic setting. Anger and excitement share elevated pitch and elevated energy and are not perceptually similar; the distinction resides in phonation, articulatory precision and fine timing, none of which a rate-pitch-energy specification captures. The sentence *Oh, that's great* spoken flatly and slowly is heard as sarcastic, spoken quickly and brightly as pleased, and spoken slowly with creaky phonation as resigned. The three readings differ in affect, not in prosodic setting alone.

Emotion is treated as a global attribute in essentially all of the literature: one label per utterance is the near-universal convention. Its status in VoxCPM2 on Khmer is unmeasured. The parenthetical channel would plausibly accept a description such as *(sounding angry)*, since it is the same free-text field that already carries *(speaking quickly)*, but this has not been tested.

2.4 Non-Verbal Vocalization

A non-verbal vocalization is a sound produced by a speaker that is not a word: laughter, a sigh, an audible breath, a filled pause, a cough, a gasp, a sob, a hesitation particle. Such sounds carry stance, regulate turn-taking and convey affect, and their presence is a substantial part of what distinguishes conversational speech from read speech.

Table 6. Non-verbal vocalization types. Tags in the first five rows are documented in VoxCPM2; the sixth row is not.

Type	Tag	Communicative function
Laughter	`[laughing]`	amusement, affiliation, mitigation
Sigh	`[sigh]`	resignation, fatigue, relief
Filled pause	`[Uhm]`	planning, hesitation, floor-holding
Attention marker	`[Shh]`	silencing, conspiratorial framing
Discourse particle	`[Question-ah]`, `[Surprise-wa]`, `[Dissatisfaction-hnn]`	stance, back-channelling, question marking
Breath, gasp, cough, sob	—	phrasing, surprise, distress, physical state

In the utterance *I thought it was fine* `[laughing]` *but apparently not*, the laugh occurs at one place, occupies a few hundred milliseconds, and leaves the speech before and after it unaffected. In *So we should* `[Uhm]` *probably wait*, a filled pause is inserted mid-clause and has the same bounded character.

Non-verbal vocalization is therefore a local event, and this is its significant property for present purposes. A global attribute competes for influence over frames that the surrounding acoustic context already predicts. A tagged event has no such competitor: the frames it occupies are predicted by nothing else, and the conditioning signal is the only available explanation for them. The engineering problem is correspondingly different, and, as Section 3.5 argues, easier.

VoxCPM2 supplies the tag inventory. Whether the tags are realised on Khmer text is unmeasured.

2.5 Related Categories

Three further categories are named here so that they are not conflated with the preceding three.

Table 7. Categories adjacent to prosody, emotion and non-verbal vocalization.

Category	Definition	Scope	Example
Voice quality	mode of phonation	global	whispered, breathy, creaky, tense
Emphasis	placement of contrastive stress	local span	*I* never said that; I never said *that*
Timing	insertion and duration of pauses	local	a deliberate pause before a resolution

Voice quality behaves as emotion does: it is global, and in principle addressable through the same descriptive channel. Emphasis and timing are local, as non-verbal vocalization is, but they require a syntax that delimits a span rather than marking a point. VoxCPM2 provides no such syntax, and adding one is a larger undertaking than adding a tag.

3 Literature Review

Four bodies of work bear on the categories set out above: natural language control of prosody and style, emotional speech synthesis, non-verbal vocalization, and the evaluation of controlled speech. Each is reviewed in turn, and Section 3.5 states which category this project will pursue and on what grounds.

3.1 Natural Language Style Control

The organising idea of this literature is to replace the reference recording with a description, and to obtain the descriptions by automatic means rather than by annotation.

Guo et al. (2023) established the format with PromptTTS, which takes a style prompt and a content prompt, encodes them separately, and conditions an acoustic model on both. The approach was constrained by its data: the style-annotated corpus had to be constructed manually, which limited its scale. Yang et al. (2023) relaxed the input format in InstructTTS, accepting free-form instructions rather than attribute lists and learning a cross-modal representation that aligns instruction text with speech style, decoded in a discrete latent space.

Leng et al. (2024) addressed the two limitations that constrain the family as a whole. The first is that a description underdetermines a voice: many distinct voices satisfy *a young woman speaking quickly*, and a model trained to map descriptions to speech must resolve that ambiguity somehow. PromptTTS 2 introduces a variation network that predicts, from the prompt representation, the reference-speech representation that would otherwise have been supplied. The second is annotation cost, addressed by a pipeline in which a speech understanding model recognises attributes and a large language model writes the corresponding prompt sentence. The system was trained on 44,000 hours.

Lyth and King (2024) give the clearest statement of the annotation argument and the one released without restriction, as Parler-TTS. Gender, accent, pitch, speaking rate and recording conditions are labelled computationally across a 45,000-hour corpus of found data using classifiers and signal measurements, and the resulting attributes are rendered as descriptive sentences on which the model is conditioned. The system outperforms prior work on fidelity while using no manually annotated data. The transferable result is that the labels required for prosodic control can be measured rather than annotated, which is the property this project relied upon when it labelled a Khmer corpus by measuring fundamental frequency, character rate and root-mean-square level per clip. Ji et al. (2024) supply the corpus counterpart in TextrolSpeech: 236 hours and 33,000 utterances with style descriptions generated by a language-model pipeline over five attribute dimensions.

This literature accounts for the parenthetical mechanism in VoxCPM2, which uses the same conditioning format and was presumably trained in the same manner. It also sets the expectation against which Section 1.4 should be read: these systems control pitch, rate and energy from description, and those are precisely the axes the measurement finds already functioning on Khmer.

3.2 Emotional Speech Synthesis

Work on emotion is organised around acted, parallel corpora. Zhou et al. (2022) survey the field and introduce the Emotional Speech Dataset, which remains the standard reference: 350 parallel utterances from ten English and ten Mandarin speakers across five emotion categories, exceeding 29 hours recorded under controlled acoustic conditions, and designed to support multi-speaker and cross-lingual conversion.

Hsu et al. (2024) are the bridge between this section and the next. Their system controls speaker emotion and laughter within a single flow-matching zero-shot model, and in doing so treats a laugh as a controllable event rather than as an emotional label. The separation between emotion and non-verbal vocalization, clear enough as a definition, is not clean in practice.

The obstacle in this category is the shape of the data rather than the adequacy of the method. Every result rests on parallel, acted, utterance-labelled emotional speech. No Khmer corpus of that description exists, and commissioning one is not proportionate to the value it would return.

3.3 Non-Verbal Vocalization

This is the most active of the four areas and the one whose methods transfer most directly to VoxCPM2.

NVSpeech (2025) is the closest match to the problem stated in Section 2.4. It treats recognition and synthesis as one pipeline. A manually annotated set of 48,430 utterances covering eighteen word-level paralinguistic categories is used to train a paralinguistic-aware speech recogniser that emits the cues as inline decodable tokens, so that a transcript reads *You're so funny [Laughter]*. That recogniser then labels a corpus of 174,179 Chinese utterances, 573 hours, with word-level alignment. A zero-shot synthesiser is finally fine-tuned on the combined human- and machine-labelled data, yielding explicit control over vocalizations inserted at arbitrary token positions. Three elements transfer: the tag is placed inline at the position of the event rather than in a header; a recogniser-shaped detector can bootstrap a corpus from found audio; and a modest human-validated seed set suffices to bootstrap the automatic labeller.

Kanda et al. (2024) provide the closest methodological reference. ELaTE fine-tunes a conditional flow-matching zero-shot synthesiser using frame-level conditioning derived from a laughter detector, obtaining control over both the timing of a laugh and its acoustic character. Two of its results constrain any plan built on it. A comparatively small conditioned dataset is sufficient; and mixing the conditioned data with general training data preserves the quality of the base model, so the fine-tune need not be paid for in intelligibility. The relevance is direct: the decoder ELaTE modifies belongs to the same class as the VoxCPM2 local diffusion transformer.

NonverbalTTS (2025) is the reference for corpus construction at a tractable scale. Seventeen hours covering ten non-verbal types were assembled from open sources by automatic detection followed by human validation, and the resulting system is reported at parity with proprietary alternatives. The detectors on which such pipelines depend are themselves available: Gillick et al. (2021) release robust frame-level laughter detection and segmentation trained on found audio, and Gong et al. (2022) release VocalSound, 21,000 crowdsourced recordings of laughter, sighs, coughs, throat-clearing, sneezes and sniffs from 3,365 speakers, together with a classifier baseline. Between them the automatic detection stage requires no model training. Where found audio is too thin to support detection, deliberate recording remains viable; MNV-17 (2025) demonstrates this for Mandarin.

Table 8. Principal references for non-verbal vocalization, by pipeline stage.

Stage	Reference	Contribution
Detection	Gillick et al. (2021); Gong et al. (2022)	Released frame-level laughter detector; released classifier over six vocalization types.
Corpus construction	NVSpeech (2025); NonverbalTTS (2025)	Recogniser-driven auto-labelling at 573 hours; detection plus human validation at 17 hours.
Synthesis	Kanda et al. (2024); Hsu et al. (2024)	Flow-matching fine-tuning with detector conditioning; joint emotion and laughter control.
Deliberate recording	MNV-17 (2025)	Performative corpus where found audio is insufficient.

Taken together these constitute a complete and published procedure: detection, alignment, human validation, an inline-tagged manifest, and a flow-matching fine-tune with general data mixed in. Every stage has either a reference implementation or a released model.

3.4 Evaluation of Controlled Speech

A control claim requires an instrument, and this project has already established that the usual instruments are unsuitable for Khmer. Across 400 clips the rank correlation between UTMOS, a learned mean-opinion-score predictor, and Khmer character error rate is +0.55: the recordings the predictor scores highest are those that render the Khmer least correctly. The inversion is between models rather than within any one model's output, which is precisely the comparison the predictor was being used to make. Naturalness predictors trained without Khmer in view cannot be relied upon here.

Two recent contributions address the evaluation of non-verbal vocalization specifically. NVV-SuperBench (2026) pairs a unified taxonomy of 45 vocalization types with a bilingual English and Chinese dataset and, more usefully, defines a protocol that separates general speech naturalness from vocalization-specific controllability, placement and salience; fifteen systems are evaluated under it. Those four axes are the appropriate ones to report against, since they decompose the question into whether the event occurred, whether it was of the requested type, whether it occurred in the requested place, and whether it was acoustically convincing.

NVMOS (2026) supplies the last of these as a model rather than as a listening panel. It predicts a mean-opinion-score-like value between zero and five for a specific marked non-verbal event, taking as input the audio together with text containing an explicit tag such as `[laugh]`. The authors additionally report that general-purpose audio-capable multimodal models disagree measurably with expert raters on this task, so a multimodal model is not an acceptable substitute. Its input format, tagged text paired with audio, is the format in which a training manifest for this work would already exist.

An evaluation plan that avoids the inverted predictors therefore exists: controllability and placement measured automatically with a detector, acoustic quality of the event measured with NVMOS, and intelligibility regression measured with the project's existing Khmer connectionist temporal classification scorer against the frozen evaluation set.

3.5 Research Focus

This project will pursue non-verbal vocalization. The reasoning proceeds by elimination and is then stated positively.

Prosody is already provided. The parenthetical mechanism moves Khmer pitch across 106.37 hertz at a rank correlation of +0.759, roughly four times the 28.27 hertz separation the project's own labelled corpus is able to express (Section 1.4). A prosodic controller trained on that corpus would reproduce a capability the base model possesses, in a conditioning channel that is already occupied. The residue, principally pitch variation and reproducibility across random seeds, is genuine but small.

Emotion is blocked on data rather than on method. The results reviewed in Section 3.2 depend without exception on acted, parallel, utterance-labelled emotional speech, and no Khmer corpus of that description exists.

Non-verbal vocalization is the remaining category, and four considerations recommend it.

(1) It is a local event. A tagged vocalization is the only predictor of the frames it occupies, whereas a global attribute must compete for influence over frames the surrounding context already determines. The conditioning difficulty that attends global-attribute training is therefore not expected to arise.

(2) The interface exists. The tags `[laughing]`, `[sigh]` and `[Uhm]` are documented in the unmodified model. The work is to make them operate on Khmer, not to design a syntax and persuade the model to read it.

(3) The acoustics are substantially language-independent. Laughter and sighing are not language-specific gestures; what is language-specific is where they are placed and what they signal in context, and placement is supplied by the tag position. This is why a small Khmer corpus may be sufficient, and it is the assumption that any methodology must test before committing resources.

(4) Every stage has a published reference: NVSpeech for the pipeline and the inline-token format, ELaTE for the flow-matching fine-tune and the data-mixing ratio, NonverbalTTS for the human-validation loop, Gillick et al. and VocalSound for detection, and NVV-SuperBench and NVMOS for evaluation.

4 Methodology

Reserved.

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Appendix A Overfitting Test of Tag Conditioning

This appendix reports a single experiment, carried out after Section 3.5 selected non-verbal vocalization as the research focus. Its purpose is diagnostic rather than demonstrative: it establishes that the model is capable of the conditioning the methodology depends on, and it does so before any effort is committed to building a corpus at scale. The experiment is conducted in English. Section 1.4 records that Khmer text reaches the model as byte fallback at roughly three tokens per character, which makes a Latin tag a conspicuously low-entropy island in the sequence; running the test in English removes that asymmetry from the result and leaves the question of Khmer transfer to be settled separately.

A.1 Motivation

A conventional fine-tune had already been performed on 3.51 hours of tagged English speech derived from deepvk/NonverbalTTS, and evaluated on 150 clips held out of training, using a low-rank adaptation of the backbone language model, the local diffusion transformer and the projection layers. Its effect on the inline tag was close to nothing. Deleting the tag from the transcript raised the flow-matching loss on 82 of 150 clips, which a sign test cannot separate from chance ($p = 0.144$), and relocating the tag to the far end of the sentence was similarly without effect ($p = 0.284$). On the unmodified model the corresponding relocation test gives $p = 0.98$, which is to say the model behaves as though the tag's position had not been changed at all.

That null result admits two explanations which lead to opposite decisions, and the audio cannot distinguish them, because a generated waveform conflates a dependence that was never learned with one that was learned and then removed by classifier-free guidance and the flow-matching solver. The distinction has to be drawn at the training objective.

(1) **The mechanism is absent.** Text conditioning reaches the diffusion head as an utterance-level vector, and nothing in the architecture carries the information that an event belongs at one particular moment. If this is the case, inline tags are not achievable in this model at any corpus size, and the work should fall back to global descriptors of the kind Section 1.4 measures.

(2) **The signal is too weak.** The mechanism exists, but 98 instances of the most common non-laughter tag, distributed over 3.51 hours, is too sparse a gradient for the objective to locate. If this is the case the approach is sound and the corpus is the thing to change.

Deliberate overfitting separates the two, and is the least expensive decisive experiment available. A small set of recordings is memorized, and the tag's influence is then measured on those same recordings. A model that fails to acquire the dependence when it has been given the answer several hundred times has the first problem. A model that acquires it has the second, and the second is tractable.

A.2 Design

Every choice in Table 9 is the reverse of what a production fine-tune would use, deliberately. The mixture of untagged material that ELaTE prescribes at a one-to-one ratio, and that Section 3.5 adopts, is omitted: its function is to prevent the base model regressing, which is a concern about generalization that this run does not have. Weight decay is zero, the learning rate is five times the documented value, and the validation manifest is the training manifest, so that the validation loss reads memorization rather than generalization.

Table 9. Configuration of the overfitting run. The full file is `finetune/conf/nvv_overfit.yaml`.

Quantity	Value	Rationale
Clips	60 (5.4 min)	12 per tag; small enough to memorize in about an hour
Tags	5	four undocumented, one documented as control
Untagged mixture	none	regularization is not wanted here
Learning rate	0.0005	five times the documented rate for this adapter
Weight decay	0	no regularization, deliberately
Warmup	50 steps	short, for the same reason
Steps	2,000	about 267 passes over the same clips
Adapter rank	64 (alpha 64)	language model, diffusion head and projections all adapted
Validation set	identical to training	validation loss reads memorization by design

Clips were admitted only if they carried exactly one tag type, occurring exactly once, in a recording of two to eight seconds. A clip containing two events teaches less per gradient step, and the objective of this run is the cleanest available mapping from one string to one event. Four of the five tags used — `[cough]`, `[groan]`, `[throat-clear]`, `[sniff]` — appear nowhere in the inventory OpenBMB publishes, reproduced in Table 6. The fifth, `[laughing]`, is documented and is retained as a control: the unmodified model already produces laughter, so a result in which the documented tag trains appreciably better than the undocumented ones would indicate that the model was retrieving prior knowledge of the word rather than acquiring a new binding.

A.3 Measurement

The instrument is a teacher-forced forward pass, one per clip per condition, comparing the flow-matching loss across the four rewritings of the transcript in Table 10. Generated audio is not used to decide the question, for the reason given in A.1.

Table 10. The four conditions of the sensitivity probe.

Condition	Transcript	What it tests
True	tag inline, at the event	reference
Removed	tag deleted	whether the tag carries information at all
Moved	tag pushed to the end of the sentence	whether the tag's **position** carries information
Scrambled	word order destroyed, tags left in place	positive control

Two features of the design determine what the numbers can be read to mean. The **moved** condition is what distinguishes a local event from an utterance-level flag: a model that has learned only that a clip contains a cough scores identically when the tag is relocated, whereas a model that has learned where the cough belongs does not. The **scrambled** condition is what makes a null result interpretable; it must be expensive, and if a destroyed transcript does not raise the loss then the probe is not measuring the objective and no other row can be believed.

The diffusion timestep and the noise are reseeded from the clip index before every forward pass, so that all four conditions for a given clip are evaluated under identical sampling. The flow-matching loss is stochastic, and an unseeded comparison would report sampling variance.

A.4 Results

Table 11. Effect of each rewriting on the flow-matching loss after memorization (n = 60; reference loss 0.23070). *Worse* counts the clips on which the rewriting raised the loss; *p* is a two-sided sign test.

Condition	Mean loss	Change	Worse	p
Removed	0.34955	+0.11885	59/60	< 0.0001
Moved	0.29089	+0.06019	43/60	0.0005
Scrambled	0.84472	+0.61402	59/60	< 0.0001

The positive control raised the loss by +0.61402 on 59 of 60 clips, so the probe is measuring the objective and the other two rows can be read.

The tag now carries information. Deleting it raises the loss on 59 of 60 clips. **Its position also carries information**, which is the more consequential of the two findings: relocating the tag without deleting it raises the loss on 43 of 60 clips, at 51 per cent of the cost of deleting it, where the same condition on the unmodified model returns $p = 0.98$. The model is not classifying the utterance as one that contains a cough. It is reading where in the sentence the string sits and placing the event there.

Absolute losses are not comparable across the three models, which are evaluated on different clips at different stages of training. Table 12 therefore compares the cost of deleting the tag as a proportion of the cost of destroying the whole transcript, which is dimensionless and internal to each model.

Table 12. Salience of the inline tag relative to the transcript, across the three states of the model. The ratio is the cost of deleting the tag divided by the cost of scrambling the transcript, measured within each model.

Model state	n	Deletion p	Relocation p	Tag / transcript
Unmodified	150	0.2313	0.9796	2.3%
Fine-tuned, 3.51 h	150	0.1442	0.2839	5.8%
Overfitted, 5.4 min	60	< 0.0001	0.0005	19.4%

The two p-value columns are sign tests on the deletion and relocation conditions respectively. The overfitted figure is 3.3 times the fine-tuned one and 8.3 times the unmodified one.

Table 13. Loss under the deletion condition, by tag. The control tag is the only one OpenBMB documents.

Tag	Documented	Mean loss when deleted
`[throat-clear]`	no	0.3745
`[sniff]`	no	0.3619
`[cough]`	no	0.3618
`[laughing]`	yes	0.3269
`[groan]`	no	0.3226

The four undocumented tags are not weaker than the documented one, which argues that the effect in Table 11 is an acquired binding rather than retrieved knowledge of the English words inside the brackets.

A.5 Interpretation and Limits

The experiment establishes that VoxCPM2's architecture can bind an arbitrary bracketed string to a specific non-verbal event at a specific position in a sentence, and that a low-rank adaptation of the language model, the diffusion head and the projections is sufficient to install that binding. The first explanation offered in A.1 is therefore refused: the null result of the conventional fine-tune was a property of the corpus, not of the model. This is the finding the methodology of Section 3.5 requires, and it was obtained for the cost of an hour of training.

Two independent observations are consistent with it. Adding a tag imposes no architectural cost, because the model is tokenizer-free at the input: `[cough]` is already segmented into ordinary subword tokens with no unknown token produced, against a vocabulary of 73,440, so there is nothing to resize and no embedding to initialize. And the unmodified model emits a short interval of non-lexical audio for any bracketed string, including strings that are not words, while never pronouncing the bracketed text aloud, which suggests a general convention that brackets denote an instruction rather than knowledge of particular tag names.

The experiment establishes nothing about generalization, and must not be quoted as though it did. The evaluation is performed on the 60 sentences the model was trained on, approximately 267 times each. That is what overfitting means and it is the point of the design, but it is also its entire limitation: the run demonstrates capability and says nothing about whether the same tags behave correctly on text the model has not seen. That is a separate experiment against a held-out set, and it has not been performed.

Two consequences follow for the methodology. The first is that the loss should be weighted over the frames in which the event occurs, so that a cough of a few hundred milliseconds is not averaged away against several seconds of speech; this is the effect the overfitting run obtained by repetition alone. The second is that the corpus should supply event timestamps precise enough for such a weighting to be applied, which the annotations used here do not. Khmer remains out of scope until tag expansion has been shown to generalize in English.

Provenance. The figures in this appendix are read at build time from `finetune/results/nvv/tag_sensitivity_overfit.json` and its counterparts for the two other model states. The trained adapter itself was deleted in error after the run and has not been regenerated; the recorded measurements and the synthesized audio survive, but reproducing the audio would require repeating the training. The procedure is documented in `finetune/OVERFIT_EXPERIMENT.md`.